

IS4302 Writing Assignment 1

Group Project: AgentBL — AI-Priced eBL-Backed RWA Trade Finance

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Question 1: Centralized, decentralized, or partially decentralized?

AgentBL is *partially decentralized*. It lets an exporter pledge an electronic bill of lading (eBL) and raise cash by selling discounted RWA tokens backed by the cargo. An AI agent decides the issue price. The parts that must be trusted by parties who do not trust each other live in smart contracts on the Injective testnet. `EBLRegistry` records who holds title to the cargo, its endorsement history and whether it is already pledged. It also refuses to register the same cargo hash twice, which targets double financing (as in the 2014 Qingdao metals fraud, where the same metal was pledged to several banks). `RWAOfferingPool` enforces the offering life cycle (`Open` → `Funded` → `InTransit` → `Repaid`, or `Paused`, `Frozen` or `Liquidation`). Neither the exporter nor the platform can skip a step, and settlement requires both a payment proof and an arrival proof. Every AI pricing decision is committed to `RiskPricingOracle` with an evidence hash, a quote hash and a replay-protected decision ID, so the price an investor paid can be audited later. Everything else runs off-chain on our server: parsing and cross-checking invoices, bills of lading and insurance policies, cargo valuation, LLM-based risk scoring and investor KYC.

We split the system this way for three reasons. First, LLM inference is costly and non-deterministic, so validators cannot re-execute it to reach consensus; only its output and a hash of its inputs belong on-chain. Second, trade documents reveal counterparties, prices and margins, so publishing them on a public ledger would leak commercial secrets. Storing only hashes lets a party later prove what was used without disclosing it. Third, tokenized trade receivables are an investment product: someone must perform KYC and be legally accountable. An eBL also has legal effect only when a reliable system issues it (e.g. Singapore’s MLETR-based Electronic Transactions Act). Hence an admin key mints eBLs and manages `InvestorComplianceGate`. The price of this design is that the owner, executor and oracle-updater keys remain points of trust. The contracts do not remove this trust, but they make every use of it visible and auditable. A production version would move these keys to a multisig shared by the carrier, the bank and the platform, and replace the single executor with an oracle (Question 2).

Question 2: Does the application need an oracle, and does AS-TREA apply?

Yes. A smart contract cannot observe ships or banks, so AgentBL needs off-chain information of three kinds:

1. **Physical events:** vessel departure (`markInTransit`), cargo arrival (`recordCargoArrival`), and port closures, typhoons and war-risk zones, which can trigger `PAUSE` or `FREEZE`.
2. **Financial facts:** whether the importer has paid (`recordImporterPayment`, which unlocks settlement), and commodity prices such as LME copper.

3. The AI risk price itself, pushed through RiskPricingOracle.

Today all three are centralized oracles, i.e. allow-listed updater and executor keys. A compromised key, or a platform colluding with an exporter, could mark a pool as repaid or keep an old high price after a war breaks out.

ASTREA applies, but only to part of the problem. It works for boolean propositions whose truth is publicly observable, so that randomly assigned voters and staked certifiers can check them independently and the honest answer becomes the focal point. Some of our triggers fit this well, for example “vessel IMO 9xxxxxx berthed at Yangshan before 30 June” or “Shanghai port suspended operations on date D ”. Anyone can check these against AIS or port-authority data, and settlement can tolerate ASTREA’s voting latency. Using ASTREA here would remove our single executor key. It does not fit the rest. (i) The AI price is a forecast, not a fact. Whether a price of 0.83 was “correct” is known only after maturity, so voters cannot verify it. Reducing it to “is risk high?” discards the information we need, and voters would simply copy the agent’s answer. What we need here is accountability (committed inputs and evidence hashes compared ex post), not truth discovery. (ii) Bank payments and document authenticity are private. Voters could only check them by seeing confidential documents, which defeats our privacy design. Signed attestations from liable parties (the issuing bank or the carrier), possibly with zero-knowledge proofs, fit better. (iii) During a crisis, repricing must take minutes, before the next subscription, and continuous prices need aggregated data feeds rather than yes/no votes. Moreover, ASTREA is secure only if certifier stakes exceed what an attacker gains, and one copper cargo is worth about USD 4.5 million. We would therefore use a hybrid: ASTREA for public binary events, bank and carrier attestations for private facts, and an accountable, evidence-committing oracle for AI pricing.